Operations during loaded and ballasted voyages;
Discharging and tank stripping: and
Emergency pro c edures , including pre-planne d action in the
event of leaks. fire, collision, stranding , emergen c y carg o
discharge. personnel casualty."
RESOLUTION MSC.33(63)
adopted on 23 May 1994
ADOPTION OF AMENDMENTS TO THE INTERNATIONAL CONVENTION
ON STANDARDS OF TRAINING, CERTIFICATION AND

---CHUNK---

SOURCE: MSC.33(63).pdf
event of leaks. fire, collision, stranding , emergen c y carg o
discharge. personnel casualty."
RESOLUTION MSC.33(63)
adopted on 23 May 1994
ADOPTION OF AMENDMENTS TO THE INTERNATIONAL CONVENTION
ON STANDARDS OF TRAINING, CERTIFICATION AND
WATCHKEEPING FOR SEAFARERS, 1978 RESOLUTION MSC.33(63)
adopted on 23 May 1994
ADOPTION OF AMENDMENTS TO THE INTERNATIONAL CONVENTION
ON STANDARDS OF TRAINING, CERTIFICATION AND
WATCHKEEPING FOR SEAFARERS, 1978

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SOURCE: MSC.34(63).pdf
RESOLUTION MSC.34(63)
adopted on 23 May 1994
ADOPTION OF AMENDMENTS TO THE CODE FOR THE
CONSTRUCTION AND EQUIPMENT OF SHIPS CARRYING
LIQUEFIED GASES IN BULK ANNEX 5
RESOLUTION MSC.34(63)
adopted on 23 May 1994
MSC 63 / 23/Add.1
ADOPTION OF AMENDMENTS TO THE CODE FOR THE CONSTRUCTION
AND EQUIPMENT OF SHIPS CARRYING LIQUEFIED GASES IN BULK
THE MARITIME SAFETY COMMITTEE,
RECALLING Article 28(b) of the Convention on the International Maritime Organization concerning the functions of the Committee,
RECALLING ALSO resolution A.328(IX), by which the Assembly, at its ninth session, adopted the Code for the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk (Gas Carrier Code) and authorized the Committee to amend the Code as may be necessary,
NOTING resolution MSC.32(63), by which the Committee adopted amendments to the International Code for the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk (IGC Code),
RECOGNIZING the need to bring the corresponding amendments to the Gas Carrier Code into force on the date on which the amendments to the IGC Code enter into force,

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SOURCE: MSC.34(63).pdf
Liquefied Gases in Bulk (IGC Code),
RECOGNIZING the need to bring the corresponding amendments to the Gas Carrier Code into force on the date on which the amendments to the IGC Code enter into force,
HAVING CONSIDERED at its sixty-third session amendments to the Gas Carrier Code proposed by the Sub-Committee on Bulk Chemicals,
1. ADOPTS amendments to the Gas Carrier Code, the text of which is given in the Annex to the present resolution;
2. DETERMINES that the amendments should become effective on 1 July 1998 upon acceptance and entry into force of the corresponding amendments to the IGC Code adopted by resolution MSC.32(63).
W/ 1648n / EWP
RESOLUTION MSC.34(63)
adopted on 23 May 1994
ADOPTION OF AMENDMENTS TO THE CODE FOR THE CONSTRUCTION AND EQUIPMENT OF SHIPS CARRYING LIQUEFIED GASES IN BULK MSC 63/23/Add.1
ANNEX 5
Page 2
ANNEX
AMENDMENTS TO THE CODE FOR THE CONSTRUCTION AND EQUIPMENT OF SHIPS CARRYING LIQUEFIED GASES IN BULK
1 Existing chapter XV is replaced by the following:
"CHAPTER XV
FILLING LIMITS FOR CARGO TANKS
15.1 General
15.1.1 No cargo tanks should have a higher filling limit (FL) than 98% at

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SOURCE: MSC.34(63).pdf
SHIPS CARRYING LIQUEFIED GASES IN BULK
1 Existing chapter XV is replaced by the following:
"CHAPTER XV
FILLING LIMITS FOR CARGO TANKS
15.1 General
15.1.1 No cargo tanks should have a higher filling limit (FL) than 98% at
the reference temperature, except as permitted by 15.1.3.
15.1.2 The maximum loading limit (LL) to which a cargo tank may be loaded should be determined by the following formula:
LL
where:
LL =
FL =
=
=
FL
loading limit expressed in percent which means the maximum allowable liquid volume relative to the tank volume to which the tank may be loaded;
filling limit as specified in 15.1.1 or 15.1.3;
relative density of cargo at the reference temperature; and

relative density of cargo at the loading temperature
and pressure.
15.1.3 The Administration may allow a higher filling limit (FL) than the
limit of 98% specified in 15.1.1 at the reference temperature, taking into
account the shape of the tank, arrangements of pressure relief valves,
accuracy of level and temperature gauging and the difference between the
loading temperature and the temperature corresponding to the vapour pressure

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SOURCE: MSC.34(63).pdf
account the shape of the tank, arrangements of pressure relief valves,
accuracy of level and temperature gauging and the difference between the
loading temperature and the temperature corresponding to the vapour pressure
of the cargo at the set pressure of the pressure relief valves, provided the
conditions specified in 8.2.17 are maintained.
15.1.4 For the purposes of this chapter only, "reference temperature" means:
(a) the temperature corresponding to the vapour pressure of the cargo at
the set pressure of the pressure relief valves when no cargo vapour
pressure/temperature control as referred to in chapter VII is
provided;
W/1648n/EWP
RESOLUTION MSC.34(63)
adopted on 23 May 1994
ADOPTION OF AMENDMENTS TO THE CODE FOR THE
CONSTRUCTION AND EQUIPMENT OF SHIPS CARRYING
LIQUEFIED GASES IN BULK MSC 63/23/Add.1
ANNEX 5
Page 3
(b) the temperature of the cargo upon termination of loading, during '
transport, or at unloading, whichever is the greatest, when a cargo
vapour pressure/temperature control as referred to in chapter VII is
provided. If this reference temperature would result in the cargo

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SOURCE: MSC.34(63).pdf
transport, or at unloading, whichever is the greatest, when a cargo
vapour pressure/temperature control as referred to in chapter VII is
provided. If this reference temperature would result in the cargo
tank becoming liquid full before the cargo reaches a temperature
corresponding to the vapour pressure of the cargo at the set
pressure of the relief valves required in 8.2, an additional
pressure relieving system complying with 8.3 should be fitted.
15.1.5 The Administration may allow type C tanks to be loaded according to
the following formula provided that the tank vent system has been approved in
accordance with 8.2.18:
LL =
where:
LL =
FL =
=
PL
PR

FL
PL
loading limit as specified in 15.1.2;
filling limit as specified in 15.1.1 or 15.1.3;
relative density of cargo at the highest temperature
which the cargo may reach upon termination of loading,
during transport, or at unloading, under the ambient
design temperature conditions described in 7.1.2; and
as specified in 15.1.2.
This paragraph does not apply to products requiring a type IG ship.
15.2 Information to be provided to the master

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SOURCE: MSC.34(63).pdf
during transport, or at unloading, under the ambient
design temperature conditions described in 7.1.2; and
as specified in 15.1.2.
This paragraph does not apply to products requiring a type IG ship.
15.2 Information to be provided to the master
The maximum allowable loading limits for each cargo tank should be
indicated for each product which may be carried, for each loading temperature
which may be applied and for the applicable maximum reference temperature, on